

Technical Documentation: Distributed Caching System

Overview:

This document describes a simplified distributed caching system designed for high-throughput web applications. The system improves latency by storing frequently accessed data closer to application servers.

Architecture:

The system consists of three main components: cache nodes, a coordinator service, and client SDKs. Cache nodes store key-value pairs in memory with optional disk persistence.

Data Flow:

1. A client requests a resource.
2. The request is checked against the nearest cache node.
3. On a cache miss, the coordinator fetches data from the origin database.
4. The result is stored back in the cache for future requests.

Consistency Model:

The system uses eventual consistency. Updates propagate asynchronously between cache nodes. This design prioritizes availability and partition tolerance.

Failure Handling:

If a cache node fails, requests are routed to the next available node. The coordinator periodically performs health checks and rebuilds failed nodes.